

- (ii) Let m be the mass of ice formed.
 M is the initial mass of water = 120 g

Heat loss forming ice = Heat removed by evaporation [C1]

$$mL_f = (M - m)L_v$$

$$m = \frac{L_v}{L_f + L_v} M$$

$$= \frac{2.52 \times 10^6}{(25.2 + 3.40) \times 10^5} (120) \quad [\text{M1}]$$

$$= 105.7 \approx 106 \text{ g} \quad [\text{A1}]$$

- 6 (a) (i) An incident highly energetic electron stops abruptly [B1] in a single collision with the target/metal nucleus.

(ii) $\frac{hc}{\lambda_{\min}} = eV$

$$V = \frac{hc}{e\lambda_{\min}} = \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{(1.6 \times 10^{-19})(0.068 \times 10^{-9})} \quad [\text{M1}]$$

$$= 18281 = 18300 \text{ V} \quad [\text{A1}]$$

(b) (i) Arrow from L to K shell, pointing downwards [B1]

(ii) $\Delta E = \frac{hc}{\lambda}$

$$\lambda = \frac{hc}{\Delta E} = \frac{(6.63 \times 10^{-34})(3.0 \times 10^8)}{(20 - 3.1)(10^3)(1.6 \times 10^{-19})} \quad [\text{M1}]$$

$$= 7.35577 \times 10^{-11} = 7.4 \times 10^{-11} \text{ m} \quad [\text{A1}]$$

(iii) To produce K_{α} characteristic X-ray, electron in the K-shell needs to be removed/ejected out of the atom. This requires energy of 20 keV. [M1]

Accelerating voltage of X-ray machine is only 18.3 kV which is less than 20kV [M1]. Hence incoming/colliding electrons (from cathode) do not have sufficient energy to eject the inner shell electron.

K_{α} characteristic X-ray cannot be produced. [A1]

(c) (i) Order of magnitude = 10^{-10} m [B1] {accept range of 10^{-9} m to 10^{-10} m }

For significant/observable/apparent diffraction to occur (to form a pattern), the slit width (i.e the spacing of between the DNA molecules) should be approximately equal to the wavelength of the X-ray used. [B1]

(ii) For the electron to have de Broglie wavelength of 0.11 nm,

$$p = \frac{h}{\lambda}$$

$$p = \frac{h}{\lambda} = \frac{(6.63 \times 10^{-34})}{(0.11 \times 10^{-9})} \quad [\text{M1}]$$

$$= 6.0273 \times 10^{-24} = 6.0 \times 10^{-24} \text{ N s} \quad [\text{A1}]$$